



Smart Lab

***1:Software Code of Main Controller
(Board 1)***

***2:Software Code of Sensor Node
(Board2)***



1: Software Code of Main Controller (Board 1)

/*

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بورد 1 - المتحكم الرئيسي

- (التحكم من أي مكان في العالم) Blynk IoT
- MQTT (Home Assistant)
- WiFi: AP دائماً + Station mode
- واجهة ويب محلية -
- يستقبل بيانات الحساسات من بورد 2 -

التغييرات:

- لمبة 1 + لمبة 2 بحساس حركة واحد -
- بحساس حرارة واحد (بقي مروحة 2 RELAY4) مروحة 1 + مروحة 2 -
- يدوي Restart تنبيه الغاز: قفل كامل دائم لحد -
- (V11) من التطبيق PIR تفعيل/تعطيل -

المكتبات المطلوبة (Arduino Library Manager):

- Blynk by Volodymyr Shymanskyy
- PubSubClient by Nick O'Leary
- ArduinoJson by Benoit Blanchon (نسخة 6.x)
- LittleFS (مدمجة مع ESP8266)

=====

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*/

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```
// ===== Blynk Config =====

#define BLYNK_TEMPLATE_ID  "TMPL5a45qncLh"

#define BLYNK_TEMPLATE_NAME "smart lab"

#define BLYNK_AUTH_TOKEN
"5YvX43GqJOwkUYfrQWhptYIEsNWELGOH"


#define BLYNK_PRINT Serial


#include <ESP8266WiFi.h>
#include <ESP8266WebServer.h>
#include <BlynkSimpleEsp8266.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include <LittleFS.h>


// ===== Pins الريليات =====

#define RELAY1 5  // D1 - 1 لمبة
#define RELAY2 4  // D2 - 2 لمبة
#define RELAY3 0  // D3 - 1 مروحة
#define RELAY4 14 // D5 - 2 مروحة ← (LED كان)
#define RELAY5 12 // D6 - داتا شو
#define RELAY6 13 // D7 - قفل الباب

const int RELAY_COUNT = 6;

const int relayPins[RELAY_COUNT] = {RELAY1, RELAY2, RELAY3,
RELAY4, RELAY5, RELAY6};

bool relayState[RELAY_COUNT] = {false};

String relayNames[RELAY_COUNT] = {
```

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```
"لمبة 1", "لمبة 2", "مروحة 1", "مروحة 2", "داتا شو", "قفل الباب"
};

// ===== Blynk Virtual Pins =====
// V1→V6 : التحكم في الريليات
// V7 : الحرارة
// V8 : الرطوبة
// V9 : الغاز
// V10 : الحركة (LED)
// V11 : PIR تفعيل/تعطيل
// V12 : تنبيه الغاز

// ===== إعدادات الشبكة =====
String apSSID = "SmartLab";
String apPassword = "12345678";
String staSSID = "";
String staPass = "";

// ===== مفاتيح الأمان =====
const char* ACCESS_KEY = "lab2026key";
const char* SENSOR_KEY = "sensorNode2026";

// ===== MQTT =====
String mqttBroker = "broker.hivemq.com";
String mqttUser = "";
String mqttPassword = "";
int mqttPort = 1883;
String mqttPrefix = "smartlab";
```

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```
WiFiClient espClient;
PubSubClient mqttClient(espClient);
unsigned long lastMqttReconnect = 0;
bool mqttEnabled = false;

// ===== بيانات الحساسات =====

float temperature = 0;
float humidity = 0;
int gasValue = 0;
bool motionDetected = false;
bool pirReady = false;
bool sensorBoardOnline = false;
unsigned long lastSensorUpdate = 0;
const unsigned long sensorTimeout = 8000;

// ===== منطق الأتمتة =====

// --- حركة ---
unsigned long lastMotionTime = 0;
const unsigned long motionTimeout = 15000; // 15 ثانية بعد ما تختفي الحركة
bool lastMotionState = false;
bool pirEnabled = true;

// --- حرارة ---
const float tempThreshold = 30.0; // فوق 30 → شغل المروحتين

// --- غاز ---
int gasThreshold = 500;
```

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```
bool gasAlert    = false;      // يدوي Restart قفل كامل دائم لحد: true لو
int  gasHighCount = 0;
const int gasConfirmReadings = 3;

// ===== Blynk Timer =====
BlynkTimer blynkTimer;
ESP8266WebServer server(80);
bool staConnected = false;

// ===== Relay Control =====
void setRelay(int idx, bool state) {
    if (idx < 0 || idx >= RELAY_COUNT) return;
    relayState[idx] = state;
    digitalWrite(relayPins[idx], state ? LOW : HIGH);
    publishRelayStatus(idx);
    Blynk.virtualWrite(V1 + idx, state ? 1 : 0);
}

void allRelaysOff() {
    for (int i = 0; i < RELAY_COUNT; i++) setRelay(i, false);
}

// ===== Blynk Handlers =====
BLYNK_WRITE(V1) { if (!gasAlert) setRelay(0, param.asInt() == 1); }
BLYNK_WRITE(V2) { if (!gasAlert) setRelay(1, param.asInt() == 1); }
BLYNK_WRITE(V3) { if (!gasAlert) setRelay(2, param.asInt() == 1); }
BLYNK_WRITE(V4) { if (!gasAlert) setRelay(3, param.asInt() == 1); }
BLYNK_WRITE(V5) { if (!gasAlert) setRelay(4, param.asInt() == 1); }
```

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```
BLYNK_WRITE(V6) { if (!gasAlert) setRelay(5, param.asInt() == 1); }
```

```
// من التطبيق PIR تفعيل/تعطيل
```

```
BLYNK_WRITE(V11) {  
  pirEnabled = (param.asInt() == 1);  
  if (!pirEnabled) {  
    setRelay(0, false);  
    setRelay(1, false);  
  }  
}
```

```
BLYNK_CONNECTED() {  
  Serial.println("☑ Blynk متصل!");  
  for (int i = 0; i < RELAY_COUNT; i++) {  
    Blynk.virtualWrite(V1 + i, relayState[i] ? 1 : 0);  
  }  
  Blynk.virtualWrite(V11, pirEnabled ? 1 : 0);  
}
```

```
// ===== Blynk بعث بيانات الحساسات لـ =====  
=====
```

```
void sendSensorsToBlynk() {  
  if (!Blynk.connected()) return;  
  Blynk.virtualWrite(V7, temperature);  
  Blynk.virtualWrite(V8, humidity);  
  Blynk.virtualWrite(V9, gasValue);  
  Blynk.virtualWrite(V10, motionDetected ? 1 : 0);  
  Blynk.virtualWrite(V12, gasAlert ? 1 : 0);  
  if (gasAlert) {
```

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```
Blynk.logEvent("gas_alert", " ⚠️ تحذير! تم اكتشاف غاز — النظام مقفول! أعد التشغيل يدوياً");
```

```
}
```

```
}
```

```
// ===== MQTT =====
```

```
String controlTopic(int idx) { return mqttPrefix + "/control/relay" + String(idx + 1); }
```

```
String statusTopic(int idx) { return mqttPrefix + "/status/relay" + String(idx + 1); }
```

```
String controlAllTopic() { return mqttPrefix + "/control/all"; }
```

```
String statusAllTopic() { return mqttPrefix + "/status/all"; }
```

```
void publishRelayStatus(int idx) {
```

```
    if (!mqttClient.connected()) return;
```

```
    mqttClient.publish(statusTopic(idx).c_str(), relayState[idx] ? "ON" : "OFF", true);
```

```
    String json = "{";
```

```
    for (int i = 0; i < RELAY_COUNT; i++) {
```

```
        json += "\"relay\" + String(i+1) + \":\" + (relayState[i] ? \"ON\" : \"OFF\") + \",\";
```

```
        if (i < RELAY_COUNT-1) json += ",";
```

```
    }
```

```
    json += "}";
```

```
    mqttClient.publish(statusAllTopic().c_str(), json.c_str(), true);
```

```
}
```

```
void publishHADiscovery() {
```

```
    if (!mqttClient.connected()) return;
```

```
    for (int i = 0; i < RELAY_COUNT; i++) {
```

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```
String uid = "smartlab_relay" + String(i+1);
String topic = "homeassistant/switch/" + uid + "/config";
StaticJsonDocument<512> doc;
doc["name"] = relayNames[i];
doc["unique_id"] = uid;
doc["command_topic"] = controlTopic(i);
doc["state_topic"] = statusTopic(i);
doc["payload_on"] = "ON";
doc["payload_off"] = "OFF";
doc["retain"] = true;
JsonObject dev = doc.createNestedObject("device");
dev["identifiers"][0] = "smartlab_board1";
dev["name"] = "Smart Lab";
dev["manufacturer"] = "Sabry Garage";
String payload; serializeJson(doc, payload);
mqttClient.publish(topic.c_str(), payload.c_str(), true);
}
}

void mqttCallback(char* topic, byte* payload, unsigned int length) {
    String t = String(topic);
    String msg = "";
    for (unsigned int i = 0; i < length; i++) msg += (char)payload[i];
    msg.toUpperCase();
    if (t == controlAllTopic()) { if (msg == "OFF") allRelaysOff(); return; }
    for (int i = 0; i < RELAY_COUNT; i++) {
        if (t == controlTopic(i) && !gasAlert) {
            if (msg == "ON") setRelay(i, true);
        }
    }
}
```

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```
        if (msg == "OFF") setRelay(i, false);
        return;
    }
}
}

bool mqttConnect() {
    if (mqttBroker.length() < 3) return false;
    mqttClient.setServer(mqttBroker.c_str(), mqttPort);
    mqttClient.setCallback(mqttCallback);
    String clientId = "SmartLab-" + String(ESP.getChipId(), HEX);
    bool ok = mqttUser.length() > 0
        ? mqttClient.connect(clientId.c_str(), mqttUser.c_str(),
            mqttPassword.c_str())
        : mqttClient.connect(clientId.c_str());
    if (ok) {
        for (int i = 0; i < RELAY_COUNT; i++)
            mqttClient.subscribe(controlTopic(i).c_str());
        mqttClient.subscribe(controlAllTopic().c_str());
        publishHADiscovery();
        for (int i = 0; i < RELAY_COUNT; i++) publishRelayStatus(i);
        Serial.println("☑ MQTT متصل!");
    }
    return ok;
}

// ===== LittleFS =====

void saveSettings() {
    StaticJsonDocument<1024> doc;
```

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```
doc["staSSID"] = staSSID;
doc["staPass"] = staPass;
doc["apSSID"] = apSSID;
doc["apPass"] = apPassword;
doc["mqttBroker"] = mqttBroker;
doc["mqttUser"] = mqttUser;
doc["mqttPass"] = mqttPassword;
doc["mqttPort"] = mqttPort;
doc["mqttPrefix"] = mqttPrefix;

for (int i = 0; i < RELAY_COUNT; i++) doc["name" + String(i)] =
relayNames[i];

File f = LittleFS.open("/config.json", "w");
if (f) { serializeJson(doc, f); f.close(); }
}
```

```
void loadSettings() {
  if (!LittleFS.exists("/config.json")) return;
  File f = LittleFS.open("/config.json", "r");
  if (!f) return;
  StaticJsonDocument<1024> doc;
  if (deserializeJson(doc, f) == DeserializationError::Ok) {
    staSSID = doc["staSSID"] | "";
    staPass = doc["staPass"] | "";
    apSSID = doc["apSSID"] | "SmartLab";
    apPassword = doc["apPass"] | "12345678";
    mqttBroker = doc["mqttBroker"] | "broker.hivemq.com";
    mqttUser = doc["mqttUser"] | "";
    mqttPassword = doc["mqttPass"] | "";
    mqttPort = doc["mqttPort"] | 1883;
```

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```
mqttPrefix = doc["mqttPrefix"] | "smartlab";

for (int i = 0; i < RELAY_COUNT; i++) relayNames[i] = doc["name" +
String(i)] | relayNames[i];

}

f.close();
}

// ===== فحص المفاتيح =====

bool checkAccessKey() {
    if (!server.hasArg("key") || server.arg("key") != ACCESS_KEY) {
        server.send(401, "application/json", "{\"error\":\"unauthorized\"}");
        return false;
    }
    return true;
}

bool checkSensorKey() {
    if (!server.hasArg("key") || server.arg("key") != SENSOR_KEY) {
        server.send(401, "application/json", "{\"error\":\"unauthorized\"}");
        return false;
    }
    return true;
}

// ===== HTML =====

String buildMainHTML() {
    String html = R"rawhtml(
<!DOCTYPE html><html lang='ar' dir='rtl'>

<head><meta charset='UTF-8'><meta name='viewport' content='width=device-
width,initial-scale=1'>
```

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```
<title>Smart Lab</title>
```

```
<style>
```

```
*{margin:0;padding:0;box-sizing:border-box}
```

```
body{font-family:'Segoe UI',sans-serif;background:linear-gradient(135deg,#0f2027,#203a43,#2c5364);min-height:100vh;display:flex;flex-direction:column;align-items:center;padding:20px 15px}
```

```
h1{color:#fff;font-size:1.9rem;margin-bottom:4px;text-shadow:0 0 20px #00d4ff}
```

```
.subtitle{color:#00d4ff;font-size:.9rem;margin-bottom:10px;letter-spacing:2px}
```

```
.nav{display:flex;gap:8px;margin-bottom:18px;flex-wrap:wrap;justify-content:center}
```

```
.nav a{background:rgba(255,255,255,.1);color:#ccc;text-decoration:none;padding:7px 14px;border-radius:20px;font-size:.82rem;border:1px solid rgba(255,255,255,.15)}
```

```
.nav a:hover{background:#00d4ff;color:#000}
```

```
.offline-banner{display:none;background:#ff9900;color:#000;padding:10px 20px;border-radius:12px;font-size:.85rem;margin-bottom:12px;width:100%;max-width:620px;text-align:center;font-weight:bold}
```

```
.offline-banner.show{display:block}
```

```
.alert-banner{display:none;background:#ff4444;color:#fff;padding:12px 20px;border-radius:12px;font-size:1rem;margin-bottom:12px;width:100%;max-width:620px;text-align:center;font-weight:bold;animation:pulse 1s infinite}
```

```
.alert-banner.show{display:block}
```

```
@keyframes pulse{0%{box-shadow:0 0 0 #ff4444}50%{box-shadow:0 0 20px #ff4444}100%{box-shadow:0 0 0 #ff4444}}
```

```
.sensor-row{display:flex;gap:12px;width:100%;max-width:620px;margin-bottom:14px}
```

```
.sensor-card{flex:1;background:rgba(255,255,255,.07);border:1px solid rgba(255,255,255,.15);border-radius:18px;padding:16px 12px;text-align:center;backdrop-filter:blur(10px)}
```

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```
.s-icon{font-size:1.7rem;margin-bottom:6px}.s-label{color:#888;font-size:.75rem;margin-bottom:4px}

.s-value{color:#00d4ff;font-size:1.4rem;font-weight:bold}.s-value.hot{color:#ff6644}

.s-sub{color:#666;font-size:.72rem;margin-top:4px}

.pir-card{background:rgba(255,255,255,.07);border:1px solid rgba(255,255,255,.15);border-radius:18px;padding:16px 22px;width:100%;max-width:620px;display:flex;justify-content:space-between;align-items:center;margin-bottom:14px;backdrop-filter:blur(10px)}

.pir-title{color:#fff;font-size:.95rem;margin-bottom:4px}

.motion-dot{width:12px;height:12px;border-radius:50%;background:#444;display:inline-block;margin-left:7px;vertical-align:middle}

.motion-dot.active{background:#00ff88;box-shadow:0 0 10px #00ff88}

.grid{display:grid;grid-template-columns:repeat(2,1fr);gap:16px;width:100%;max-width:620px}

.card{background:rgba(255,255,255,.07);border:1px solid rgba(255,255,255,.15);border-radius:18px;padding:22px 12px;text-align:center;backdrop-filter:blur(10px);transition:transform .2s}

.card:hover{transform:translateY(-3px)}

.icon{font-size:2rem;margin-bottom:8px}.name{color:#ccc;font-size:.95rem;margin-bottom:14px}

.toggle{width:62px;height:32px;background:#444;border-radius:32px;position:relative;cursor:pointer;margin:0 auto;border:none;transition:background .3s}

.toggle.on{background:#00d4ff;box-shadow:0 0 12px #00d4ff88}

.toggle.auto{background:#ff9900;box-shadow:0 0 12px #ff990088}

.toggle::after{content:"";width:24px;height:24px;background:#fff;border-radius:50%;position:absolute;top:4px;left:4px;transition:left .3s}

.toggle.on::after,.toggle.auto::after{left:34px}

.toggle.disabled{opacity:.4;cursor:not-allowed}

.status{color:#888;font-size:.78rem;margin-top:8px}.status.on{color:#00d4ff}.status.auto{color:#ff9900}
```

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```
.blynk-badge{background:rgba(0,212,100,.15);border:1px solid
#00d464;border-radius:10px;padding:6px 14px;color:#00d464;font-
size:.78rem;margin-bottom:14px;text-align:center;width:100%;max-
width:620px}

.footer{color:#555;margin-top:30px;font-size:.75rem;text-align:center}

</style></head><body>

<h1> ⚡ Smart Lab</h1>

<div class='subtitle'>لوحة التحكم الذكية</div>

<div class='nav'>

  <a href='/'> 🏠 الرئيسية </a>

  <a href='/wifi'> 📶 WiFi</a>

  <a href='/mqtt'> 📡 MQTT</a>

  <a href='/relaynames'> ✎ الأسماء</a>

  <a href='/restart'> 🔄 إعادة تشغيل</a>

</div>

<div class='blynk-badge'> 📶 Blynk IoT — تحكم من أي مكان في العالم</div>

<div class='offline-banner' id='offlineBanner'> ⚠️ <div class='alert-banner' id='alertBanner'> ⚡ غاز مكتشف! النظام مقفول بالكامل — 🛑 Restart أعد التشغيل يدوياً من زر</div>

<div class='sensor-row'>

  <div class='sensor-card'><div class='s-icon'> 🌡️ </div><div class='s-
label'>الحرارة</div><div class='s-value' id='tempVal'>--</div><div class='s-
sub'>30°C</div></div>

  <div class='sensor-card'><div class='s-icon'> 💧 </div><div class='s-
label'>الرطوبة</div><div class='s-value' id='humVal'>--</div><div class='s-
sub'>DHT11</div></div>

  <div class='sensor-card'><div class='s-icon'> 🌫️ </div><div class='s-
label'>الغاز</div><div class='s-value' id='gasVal'>--</div><div class='s-
sub'>MQ-2</div></div>

</div>
```

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```
<div class='pir-card'>

  <div><div class='pir-title'>  حساس الحركة PIR <span class='motion-dot'
id='motionDot'></span></div>

  <span id='pirStatus' style='font-size:.82rem;color:#ccc'> جاري
القراءة...</span></div>

  <button class='toggle 'rawhtml";

  html += pirEnabled ? "on" : "";

  html += R"rawhtml(' id='pirToggle' onclick='togglePIR()')></button>

</div>

<div class='grid' id='grid'></div>

<div class='footer'>Smart Lab — ESP8266 + Blynk IoT | Sabry Garage</div>

<script>

const ACCESS_KEY=')rawhtml";

  html += ACCESS_KEY;

  html += R"rawhtml(';

const relays=[]rawhtml";

  for (int i = 0; i < RELAY_COUNT; i++) {

    html += "{id:" + String(i+1) + ",name:" + relayNames[i] + ",state:" +
(relayState[i]?"true":"false") + "}";

    if (i < RELAY_COUNT-1) html += ",";

  }

  html += R"rawhtml(';

const icons=['💡','💡','🌀','🌀','🏠','🔒'];

let fanAuto=false,gasAlertActive=false;

function buildCards(){

  const g=document.getElementById('grid');g.innerHTML=";

  relays.forEach((r,i)=>{

    // كلتيهما تشتغل تلقائياً بالحرارة (i=3) ومروحة 2 (i=2) مروحة 1

    const isFan=(i===2||i===3);
```

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```
const cls=isFan&&fanAuto?'auto':(r.state?'on:');
const st=isFan&&fanAuto?' تلقائي:(r.state?'☒ واقف  ':('شغال'));
const sc=isFan&&fanAuto?'auto':(r.state?'on:');

g.innerHTML+=`<div class='card'><div class='icon'>${icons[i]}</div><div
class='name'>${r.name}</div>

  <button class='toggle ${cls}' onclick='toggle(${r.id})'
  ${gasAlertActive?"disabled":""}></button>

  <div class='status ${sc}'>${st}</div></div>`;

});

}

function toggle(id){
  if(gasAlertActive)return;

  fetch('/toggle?relay='+id+'&key='+ACCESS_KEY).then(r=>r.json()).then(d=>
  {
    if(!d.error){relays[id-1].state=d.state;buildCards();}
  });
}

function togglePIR(){
  if(gasAlertActive)return;

  fetch('/pirToggle?key='+ACCESS_KEY).then(r=>r.json()).then(d=>{
    if(!d.error){document.getElementById('pirToggle').className='toggle
'+(d.enabled?'on:');}
  });
}

setInterval(()=>{
  fetch('/status').then(r=>r.json()).then(d=>{

document.getElementById('offlineBanner').className=d.sensorOnline?'offline
-banner':'offline-banner show';

    gasAlertActive=d.gasAlert;
```

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```
document.getElementById('alertBanner').className=d.gasAlert?'alert-  
banner show':'alert-banner';  
  
const  
te=document.getElementById('tempVal'),he=document.getElementById('hum  
Val'),ge=document.getElementById('gasVal');  
  
te.textContent=d.temp+'°C';he.textContent=d.hum+'%';ge.textContent=d.gasVa  
lue;  
  
te.className=d.temp>=30?'s-value hot':'s-value';  
  
ge.className=d.gasAlert?'s-value hot':'s-value';  
  
fanAuto=d.fanAuto;  
  
const  
dot=document.getElementById('motionDot'),ps=document.getElementById('pi  
rStatus');  
  
document.getElementById('pirToggle').disabled=gasAlertActive;  
  
if(!d.pirEnabled){dot.className='motion-dot';ps.textContent='⊘ معطل';}  
  
else if(!d.pirReady){dot.className='motion-dot';ps.textContent='⌚ الحساس  
بيستقر...!';}  
  
else{dot.className='motion-dot  
'+(d.motion?'active:');ps.textContent=d.motion?'⚡ حركة مكتشفة! لا توجد  
حركة';}  
  
d.relays.forEach((s,i)=>{relays[i].state=s;});  
  
buildCards();  
  
});  
,1000);  
buildCards();  
</script></body></html>)rawhtml";  
  
return html;  
}
```

```
String buildWifiHTML() {
```

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```
bool isConnected = (WiFi.status() == WL_CONNECTED);
```

```
String html = "<!DOCTYPE html><html lang='ar' dir='rtl'><head><meta  
charset='UTF-8'><meta name='viewport' content='width=device-width,initial-  
scale=1'><title>WiFi</title>";
```

```
html += "<style>*{margin:0;padding:0;box-sizing:border-box}body{font-  
family:'Segoe UI',sans-serif;background:linear-  
gradient(135deg,#0f2027,#203a43,#2c5364);min-  
height:100vh;display:flex;flex-direction:column;align-  
items:center;padding:25px 15px;color:#fff}h2{font-size:1.5rem;margin-  
bottom:5px}.nav{display:flex;gap:8px;margin:12px 0 20px;flex-  
wrap:wrap;justify-content:center}.nav  
a{background:rgba(255,255,255,.1);color:#ccc;text-  
decoration:none;padding:7px 14px;border-radius:20px;font-  
size:.82rem;border:1px solid rgba(255,255,255,.15)}.nav  
a:hover{background:#00d4ff;color:#000}.card{background:rgba(255,255,255,.  
07);border:1px solid rgba(255,255,255,.15);border-  
radius:18px;padding:22px;width:100%;max-width:500px;margin-  
bottom:18px;backdrop-filter:blur(10px)}.card h3{color:#00d4ff;margin-  
bottom:14px;font-size:1rem}.badge{padding:3px 10px;border-  
radius:10px;font-  
size:.78rem}.badge.ok{background:#00c853;color:#000}.badge.fail{backgroun  
d:#ff4444;color:#fff}.row{display:flex;justify-content:space-between;margin-  
bottom:7px;font-size:.85rem;color:#ccc}.row span:last-child{color:#fff;font-  
weight:bold}input{width:100%;background:rgba(255,255,255,.1);border:1px  
solid rgba(255,255,255,.2);border-radius:10px;padding:11px  
14px;color:#fff;font-size:.9rem;margin-  
bottom:10px}input::placeholder{color:#666}.btn{width:100%;padding:13px;b  
order:none;border-radius:12px;font-size:.95rem;cursor:pointer;font-  
weight:bold;margin-  
top:4px}.g{background:#00c853;color:#000}.r{background:#ff4444;color:#fff  
}.b{background:#2196F3;color:#fff}.o{background:#ff9900;color:#000}.note{  
color:#888;font-size:.78rem;margin-top:8px;text-  
align:center}</style></head><body>";
```

```
html += "<h2>📶 WiFi Settings</h2><div class='nav'><a href='/'>🏠</a><a  
href='/wifi'>📶 WiFi</a><a href='/mqtt'>🐼 MQTT</a><a  
href='/relaynames'>✍️ الأسماء</a></div>";
```

```
html += "<div class='card'><h3>📶 Current Status</h3>";
```

Smart --- Lab

```
html += "<div class='row'><span>Status:</span><span>" +  
String(isConnected ? "<span class='badge ok'>☑ Connected</span>" : "<span  
class='badge fail'>✗ AP Mode Only</span>") + "</span></div>";
```

```
html += "<div class='row'><span>Network:</span><span>" +  
String(isConnected ? WiFi.SSID() : "-") + "</span></div>";
```

```
html += "<div class='row'><span>STA IP:</span><span>" +  
String(isConnected ? WiFi.localIP().toString() : "-") + "</span></div></div>";
```

```
html += "<div class='card'><h3>📶 AP (Always Active)</h3><div  
class='row'><span>SSID:</span><span>" + apSSID + "</span></div><div  
class='row'><span>Password:</span><span>" + apPassword +  
"</span></div><div  
class='row'><span>IP:</span><span>192.168.4.1</span></div></div>";
```

```
html += "<div class='card'><h3>✎ Connect to WiFi</h3><input type='text'  
id='ssid' placeholder='Network name (SSID)'><input type='password' id='pass'  
placeholder='Password'><button class='btn g' onclick='saveWifi()'>💾 Save  
and restart</button><p class='note'>Blynk تلقائياً هيشغل بعد الحفظ</p></div>";
```

```
html += "<div class='card'><h3>⚙ AP Settings</h3><input type='text'  
id='apN' placeholder='AP Name' value='" + apSSID + "'><input  
type='password' id='apP' placeholder='AP Password' value='" + apPassword +  
"'><button class='btn o' onclick='saveAP()'>💾 Save AP</button><button  
class='btn r' style='margin-top:8px' onclick='clearS()'>🗑 Clear  
settings</button><button class='btn b' style='margin-top:8px'  
onclick='location.href=\"/restart\"'>🔄 Restart</button></div>";
```

```
html += "<script>function saveWifi(){const  
s=document.getElementById('ssid').value,p=document.getElementById('pass').  
value;if(!s){alert('أدخل  
SSID');return;}fetch('/saveWifi?ssid='+encodeURIComponent(s)+'&pass='+en  
codeURIComponent(p)+'&key='" + String(ACCESS_KEY) +  
"').then(r=>r.json()).then(d=>alert(d.msg));}";
```

```
html += "function saveAP(){const  
n=document.getElementById('apN').value,p=document.getElementById('apP').  
value;if(!n||p.length<8){alert('أحرف على  
الأقل');return;}fetch('/saveAP?name='+encodeURIComponent(n)+'&pass='+enc  
odeURIComponent(p)+'&key='" + String(ACCESS_KEY) +  
"').then(r=>r.json()).then(d=>alert(d.msg));}";
```

Smart --- Lab

```
html += "function clearS(){if(!confirm('هتتمسح  
؟الإعدادات'))return;fetch('/clearSettings?key=' + String(ACCESS_KEY) +  
").then(r=>r.json()).then(d=>alert(d.msg));}</script></body></html>";  
  
return html;  
}
```

```
String buildMqttHTML() {
```

```
String html = "<!DOCTYPE html><html lang='ar' dir='rtl'><head><meta  
charset='UTF-8'><meta name='viewport' content='width=device-width,initial-  
scale=1'><title>MQTT</title>";
```

```
html += "<style>*{margin:0;padding:0;box-sizing:border-box}body{font-  
family:'Segoe UI',sans-serif;background:linear-  
gradient(135deg,#0f2027,#203a43,#2c5364);min-  
height:100vh;display:flex;flex-direction:column;align-  
items:center;padding:25px 15px;color:#fff}h2{font-size:1.5rem;margin-  
bottom:5px}.nav{display:flex;gap:8px;margin:12px 0 20px;flex-  
wrap:wrap;justify-content:center}.nav  
a{background:rgba(255,255,255,.1);color:#ccc;text-  
decoration:none;padding:7px 14px;border-radius:20px;font-  
size:.82rem;border:1px solid rgba(255,255,255,.15)}.nav  
a:hover{background:#00d4ff;color:#000}.card{background:rgba(255,255,255,.  
07);border:1px solid rgba(255,255,255,.15);border-  
radius:18px;padding:22px;width:100%;max-width:500px;margin-  
bottom:18px;backdrop-filter:blur(10px)}.card h3{color:#00d4ff;margin-  
bottom:14px;font-  
size:1rem}input{width:100%;background:rgba(255,255,255,.1);border:1px  
solid rgba(255,255,255,.2);border-radius:10px;padding:11px  
14px;color:#fff;font-size:.9rem;margin-  
bottom:10px}input::placeholder{color:#666}label{color:#888;font-  
size:.82rem;display:block;margin-  
bottom:4px}.topics{background:rgba(0,0,0,.3);border-  
radius:10px;padding:12px;font-size:.78rem;color:#ccc;line-height:1.9}.topics  
span{color:#00d4ff}.btn{width:100%;padding:13px;border:none;border-  
radius:12px;font-size:.95rem;cursor:pointer;font-  
weight:bold;background:#00c853;color:#000}</style></head><body>";
```

```
html += "<h2>  MQTT Settings</h2><div class='nav'><a  
href='/'>  </a><a href='/wifi'>  WiFi</a><a href='/mqtt'>   
MQTT</a><a href='/relaynames'>  الأسماء</a></div>";
```

Smart --- Lab

```
html += "<div class='card'><h3>🔧 Configure MQTT Broker</h3>";

html += "<label>Broker</label><input type='text' id='b' value='' +
mqttBroker + \" placeholder='broker.hivemq.com'>\";

html += "<label>Username</label><input type='text' id='u' value='' +
mqttUser + \" placeholder='اتركه فاضي لو مفيش'>\";

html += "<label>Port</label><input type='number' id='p' value='' +
String(mqttPort) + \">\";

html += "<label>Prefix</label><input type='text' id='pr' value='' + mqttPrefix
+ \" placeholder='smartlab'>\";

html += "<div class='topics'>Control: <span>\" + mqttPrefix +
\"/control/relay1</span><br>Status: <span>\" + mqttPrefix +
\"/status/relay1</span><br>Control All: <span>\" + mqttPrefix +
\"/control/all</span><br>HA Discovery:
<span>homeassistant/switch/smartlab_relay1/config</span></div>\";

html += "<button class='btn' onclick='saveMqtt()' style='margin-
top:14px'>💾 Save MQTT Settings</button></div>\";

html += "<script>function saveMqtt(){const
b=document.getElementById('b').value,u=document.getElementById('u').value
,p=document.getElementById('p').value,pr=document.getElementById('pr').val
ue;fetch('/saveMqtt?broker='+encodeURIComponent(b)+'&user='+encodeURIComponent(u)+'&port='+p+'&prefix='+encodeURIComponent(pr)+'&key=\" +
String(ACCESS_KEY) +
\"').then(r=>r.json()).then(d=>alert(d.msg));}</script></body></html>\";

return html;
}
```

```
String buildNamesHTML() {
```

```
String html = "<!DOCTYPE html><html lang='ar' dir='rtl'><head><meta
charset='UTF-8'><meta name='viewport' content='width=device-width,initial-
scale=1'><title>Names</title>\";
```

```
html += "<style>*{margin:0;padding:0;box-sizing:border-box}body{font-
family:'Segoe UI',sans-serif;background:linear-
gradient(135deg,#0f2027,#203a43,#2c5364);min-
height:100vh;display:flex;flex-direction:column;align-
items:center;padding:25px 15px;color:#fff}h2{font-size:1.5rem;margin-
```

Smart --- Lab

```
bottom:5px}.nav{display:flex;gap:8px;margin:12px 0 20px;flex-
wrap:wrap;justify-content:center}.nav
a{background:rgba(255,255,255,.1);color:#ccc;text-
decoration:none;padding:7px 14px;border-radius:20px;font-
size:.82rem;border:1px solid rgba(255,255,255,.15)}.nav
a:hover{background:#00d4ff;color:#000}.card{background:rgba(255,255,255,.
07);border:1px solid rgba(255,255,255,.15);border-
radius:18px;padding:22px;width:100%;max-width:500px;backdrop-
filter:blur(10px)}.card h3{color:#00d4ff;margin-bottom:14px;font-
size:1rem}.lbl{color:#00d4ff;font-size:.82rem;margin-bottom:5px;margin-
top:10px}input{width:100%;background:rgba(255,255,255,.1);border:1px
solid rgba(255,255,255,.2);border-radius:10px;padding:11px
14px;color:#fff;font-
size:.9rem}.btn{width:100%;padding:13px;border:none;border-
radius:12px;font-size:.95rem;cursor:pointer;font-
weight:bold;background:#00c853;color:#000;margin-
top:16px}</style></head><body>";
```

```
html += "<h2> 
```

```
html += "<div class='card'><h3> 
```

```
for (int i = 0; i < RELAY_COUNT; i++) {

    html += "<div class='lbl'>• Relay " + String(i+1) + "</div><input
type='text' id='r' + String(i) + " value='" + relayNames[i] + "'>";

}
```

```
html += "<button class='btn' onclick='saveN()'> 
```

```
html += "<script>function saveN(){let url='/saveNames?key=' +
String(ACCESS_KEY) + '';}for(let
i=0;i<6;i++)url+='&n'+i+'='+encodeURIComponent(document.getElementById('r'+i).value);fetch(url).then(r=>r.json()).then(d=>alert(d.msg));}</script></bo
dy></html>";
```

```
return html;
```

```
}
```

```
// ===== Routes =====
```

Smart --- --- Lab

```
void handleRoot() { server.send(200, "text/html; charset=UTF-8",
buildMainHTML()); }

void handleWifi() { server.send(200, "text/html; charset=UTF-8",
buildWifiHTML()); }

void handleMqtt() { server.send(200, "text/html; charset=UTF-8",
buildMqttHTML()); }

void handleNames() { server.send(200, "text/html; charset=UTF-8",
buildNamesHTML()); }
```

```
void handleRestart() {
    server.send(200, "text/html; charset=UTF-8",
        "<html><body style='background:#0f2027;color:#fff;text-align:center;padding:50px;font-family:sans-serif'>"
        "<h2>🔄 إعادة التشغيل...</h2></body></html>");
    delay(1500); ESP.restart();
}
```

```
void handleToggle() {
    if (!checkAccessKey()) return;

    if (gasAlert) { server.send(403, "application/json", "{\"error\":\"gas alert - restart required\"}"); return; }

    if (server.hasArg("relay")) {
        int idx = server.arg("relay").toInt() - 1;
        if (idx >= 0 && idx < RELAY_COUNT) {
            setRelay(idx, !relayState[idx]);
            server.send(200, "application/json",
                "{\"relay\":\"" + String(idx+1) + "\",\"state\":\"" +
                (relayState[idx]?"true":"false") + "\"}");
            return;
        }
    }
}
```

Smart --- Lab

```
server.send(400, "application/json", "{\"error\":\"invalid\"}");  
}
```

```
void handlePIRToggle() {  
  if (!checkAccessKey()) return;  
  pirEnabled = !pirEnabled;  
  if (!pirEnabled) {  
    setRelay(0, false);  
    setRelay(1, false);  
  }  
  Blynk.virtualWrite(V11, pirEnabled ? 1 : 0);  
  server.send(200, "application/json", "{\"enabled\":\"" +  
String(pirEnabled?"true":"false") + "\"}");  
}
```

```
void handleSensorUpdate() {  
  if (!checkSensorKey()) return;  
  if (server.hasArg("temp")) temperature = server.arg("temp").toFloat();  
  if (server.hasArg("hum")) humidity = server.arg("hum").toFloat();  
  if (server.hasArg("gas")) gasValue = server.arg("gas").toInt();  
  if (server.hasArg("motion")) motionDetected = (server.arg("motion") == "1");  
  if (server.hasArg("ready")) pirReady = (server.arg("ready") == "1");  
  lastSensorUpdate = millis();  
  sensorBoardOnline = true;  
  server.send(200, "application/json", "{\"ok\":true}");  
}
```

```
void handleStatus() {  
  bool fanAuto = (temperature >= tempThreshold);
```

Smart --- Lab

```
String json = "{\"gasValue\":\"" + String(gasValue) +
    "\",\"gasAlert\":\"" + (gasAlert?"true":"false") +
    "\",\"temp\":\"" + String(temperature, 1) +
    "\",\"hum\":\"" + String(humidity, 1) +
    "\",\"fanAuto\":\"" + (fanAuto?"true":"false") +
    "\",\"motion\":\"" + (motionDetected?"true":"false") +
    "\",\"pirEnabled\":\"" + (pirEnabled?"true":"false") +
    "\",\"pirReady\":\"" + (pirReady?"true":"false") +
    "\",\"sensorOnline\":\"" + (sensorBoardOnline?"true":"false") +
    "\",\"relays\":[\"";
for (int i = 0; i < RELAY_COUNT; i++) {
    json += (relayState[i]?"true":"false");
    if (i < RELAY_COUNT-1) json += ",";
}
json += "]}";
server.send(200, "application/json", json);
}

void handleSaveWifi() {
    if (!checkAccessKey()) return;
    staSSID = server.arg("ssid"); staPass = server.arg("pass");
    saveSettings();
    server.send(200, "application/json", "{\"msg\":\"إعادة التشغيل - جاري الحفظ - تم\"}");
    delay(1500); ESP.restart();
}

void handleSaveAP() {
    if (!checkAccessKey()) return;
    apSSID = server.arg("name"); apPassword = server.arg("pass");
```

Smart --- Lab

```
    saveSettings();
    server.send(200, "application/json", "{\"msg\":\"تم حفظ AP\"}");
}

void handleSaveMqtt() {
    if (!checkAccessKey()) return;
    mqttBroker = server.arg("broker");
    mqttUser = server.arg("user");
    mqttPort = server.arg("port").toInt() ?: 1883;
    mqttPrefix = server.arg("prefix");
    saveSettings();
    server.send(200, "application/json", "{\"msg\":\"تم حفظ MQTT - أعد التشغيل\"}");
}

void handleSaveNames() {
    if (!checkAccessKey()) return;
    for (int i = 0; i < RELAY_COUNT; i++) {
        String k = "n" + String(i);
        if (server.hasArg(k) && server.arg(k).length() > 0) relayNames[i] =
server.arg(k);
    }
    saveSettings();
    server.send(200, "application/json", "{\"msg\":\"تم حفظ الأسماء\"}");
}

void handleClearSettings() {
    if (!checkAccessKey()) return;
    LittleFS.remove("/config.json");
    server.send(200, "application/json", "{\"msg\":\"أعد التشغيل - امسح\"}");
}

// ===== Setup =====
```

Smart --- Lab

```
void setup() {  
    Serial.begin(115200);  
  
    Serial.println("\n ⚡ Smart Lab + Blynk IoT Starting...");  
  
    if (!LittleFS.begin()) {  
        Serial.println("❌ LittleFS فشل");  
    } else {  
        loadSettings();  
        Serial.println("✅ الإعدادات اتحملت");  
    }  
  
    for (int i = 0; i < RELAY_COUNT; i++) {  
        pinMode(relayPins[i], OUTPUT);  
        digitalWrite(relayPins[i], HIGH);  
    }  
  
    WiFi.mode(WIFI_AP_STA);  
    WiFi.softAP(apSSID.c_str(), apPassword.c_str());  
    Serial.println("📶 AP: " + apSSID + " | IP: 192.168.4.1");  
  
    if (staSSID.length() > 0) {  
        Serial.println("📶 يتصل بـ: " + staSSID);  
        WiFi.begin(staSSID.c_str(), staPass.c_str());  
        int tries = 0;  
        while (WiFi.status() != WL_CONNECTED && tries < 20) {  
            delay(500); Serial.print("."); tries++;  
        }  
        if (WiFi.status() == WL_CONNECTED) {
```

Smart --- Lab

```
staConnected = true;

Serial.println("\n☑ WiFi متصل: " + WiFi.localIP().toString());

mqttEnabled = (mqttBroker.length() > 3);

Blynk.config(BLYNK_AUTH_TOKEN);

Blynk.connect(3000);

} else {

    Serial.println("\n⚠ WiFi فشل — AP mode فقط");

}

} else {

    Serial.println("⚠ SmartLab محفوظ — اتصل بـ WiFi مفيش 192.168.4.1/wifi");

}

if (mqttEnabled) mqttConnect();

blynkTimer.setInterval(3000L, sendSensorsToBlynk);

server.on("/",      handleRoot);
server.on("/wifi",   handleWifi);
server.on("/mqtt",   handleMqtt);
server.on("/relaynames", handleNames);
server.on("/restart", handleRestart);
server.on("/toggle", handleToggle);
server.on("/pirToggle", handlePIRToggle);
server.on("/sensorUpdate", handleSensorUpdate);
server.on("/status", handleStatus);
server.on("/saveWifi", handleSaveWifi);
server.on("/saveAP", handleSaveAP);
```

Smart --- Lab

```
server.on("/saveMqtt",  handleSaveMqtt);
server.on("/saveNames",  handleSaveNames);
server.on("/clearSettings",handleClearSettings);
server.begin();
Serial.println("🌐 Web Server جاهز");
}

// ===== Loop =====
void loop() {
  if (staConnected) Blynk.run();
  blynkTimer.run();
  server.handleClient();

  // MQTT reconnect
  if (mqttEnabled && !mqttClient.connected()) {
    if (millis() - lastMqttReconnect > 5000) {
      lastMqttReconnect = millis();
      mqttConnect();
    }
  }
  if (mqttClient.connected()) mqttClient.loop();

  // Sensor board timeout
  if (sensorBoardOnline && millis() - lastSensorUpdate > sensorTimeout) {
    sensorBoardOnline = false;
    Serial.println("⚠️ بورد الحساسات offline");
  }
}
```

Smart --- Lab

// ===== غاز → قفل كامل دائم =====

// يدوي Restart لو الغاز اتكشف، فضّل كل حاجة ومنعها نهائياً لحد

```
if (!gasAlert) {  
  if (gasValue >= gasThreshold) {  
    if (++gasHighCount >= gasConfirmReadings) {  
      gasAlert = true;  
      allRelaysOff();  
      Serial.println("غاز مكتشف! النظام مقفول بالكامل - أعد التشغيل يدوياً لإرجاعه 🏠");  
      Blynk.logEvent("gas_alert", "غاز مكتشف! النظام مقفول - أعد التشغيل يدوياً 🏠");  
    }  
  } else {  
    gasHighCount = 0;  
  }  
} else {  
  // النظام مقفول: امنع أي تشغيل، فضّل القفل ثابت  
  allRelaysOff();  
}
```

// ===== حركة → لمبة 1 + لمبة 2 =====

=====

```
if (!gasAlert && pirEnabled && pirReady) {  
  if (motionDetected && !lastMotionState) {  
    // حركة جديدة → شغل اللمبتين  
    lastMotionTime = millis();  
    if (!relayState[0]) { setRelay(0, true); Serial.println " 🏠 حركة! لمبة 1  
    ("اشتغلت"); }  
    if (!relayState[1]) { setRelay(1, true); Serial.println " 🏠 حركة! لمبة 2  
    ("اشتغلت"); }  
  }  
}
```

Smart --- Lab

```
if (!motionDetected && lastMotionState) {  
    الحركة اختفت → ابدأ عدّ التايم أوت //  
    lastMotionTime = millis();  
}  
// لو مفيش حركة وعدى التايم أوت → طَقّي اللمبتين  
if (!motionDetected && millis() - lastMotionTime > motionTimeout) {  
    if (relayState[0]) { setRelay(0, false); Serial.println("لمبة 1 انتظفت"); }  
    if (relayState[1]) { setRelay(1, false); Serial.println("لمبة 2 انتظفت"); }  
}  
lastMotionState = motionDetected;  
}  
  
// حرارة → مروحة 1 + مروحة 2 =====  
=====  
if (!gasAlert) {  
    if (temperature >= tempThreshold) {  
        if (!relayState[2]) { setRelay(2, true); Serial.println("مروحة 1 اشتغلت"); }  
        if (!relayState[3]) { setRelay(3, true); Serial.println("مروحة 2 اشتغلت"); }  
    } else if (temperature < tempThreshold - 2) {  
        if (relayState[2]) { setRelay(2, false); Serial.println("مروحة 1 انتظفت"); }  
        if (relayState[3]) { setRelay(3, false); Serial.println("مروحة 2 انتظفت"); }  
    }  
}  
  
delay(100);  
}
```

Smart Lab

2:Software Code of Sensor Node (Board2):

/*

=====

بورد 2 - بورد الحساسات

- DHT11 (حرارة + رطوبة)

- MQ-2 (غاز)

- PIR (حركة)

- Station يتصل بشبكة بورد 1 كـ

- HTTP GET يبعث البيانات لبورد 1 كل 2 ثانية عبر

=====

المكتبات المطلوبة:

- DHT sensor library by Adafruit

- Adafruit Unified Sensor

*/

#include <ESP8266WiFi.h>

#include <ESP8266HTTPClient.h>

#include <WiFiClient.h>

#include <DHT.h>

// ===== بيانات شبكة بورد 1 =====

// من بورد 1، عدّله هنا كمان AP ملاحظة: لو غيّرت اسم

const char* ssid = "SmartLab";

const char* password = "12345678";

const char* MAIN_BOARD_IP = "192.168.4.1";

Smart --- Lab

```
const char* SENSOR_KEY    = "sensorNode2026";

// ===== Pins =====

#define GAS_PIN  A0    // MQ-2 (Analog)
#define DHT_PIN  15    // D8 - DHT11
#define DHT_TYPE DHT11
#define PIR_PIN   2    // D4 - PIR

DHT dht(DHT_PIN, DHT_TYPE);

// ===== متغيرات =====

float temperature    = 0;
float humidity       = 0;
int  gasValue        = 0;
bool motionDetected = false;

bool pirReady        = false;
unsigned long pirStartTime = 0;
const unsigned long pirWarmupTime = 30000; // 30 ثانية للاستقرار

unsigned long lastDHTRead = 0;
unsigned long lastSendTime = 0;
const unsigned long sendInterval = 2000;

int dhtFailCount = 0;

// ===== WiFi =====

void connectWiFi() {
```

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```
if (WiFi.status() == WL_CONNECTED) return;

Serial.println("  📶 بيتصل بـ SmartLab AP...");

WiFi.mode(WIFI_STA);

WiFi.begin(ssid, password);

int attempts = 0;

while (WiFi.status() != WL_CONNECTED && attempts < 30) {

    delay(500); Serial.print("."); attempts++;

}

if (WiFi.status() == WL_CONNECTED) {

    Serial.println("\n✅ اتصل! IP: " + WiFi.localIP().toString());

} else {

    Serial.println("\n❌ فشل الاتصال - سيحاول ثاني");

}

}

// ===== إرسال البيانات =====

void sendDataToMain() {

    if (WiFi.status() != WL_CONNECTED) { connectWiFi(); return; }

    WiFiClient client;

    HTTPClient http;

    String url = "http://";

    url += MAIN_BOARD_IP;

    url += "/sensorUpdate?key=";

    url += SENSOR_KEY;

    url += "&temp=" + String(temperature, 1);

    url += "&hum=" + String(humidity, 1);
```

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```
url += "&gas=" + String(gasValue);
url += "&motion=" + String(motionDetected ? "1" : "0");
url += "&ready=" + String(pirReady ? "1" : "0");

http.begin(client, url);
int code = http.GET();
if (code > 0) {
    Serial.println("📩 تم الإرسال - رد " + String(code));
} else {
    Serial.println("⚠️ فشل الإرسال: " + http.errorToString(code));
}
http.end();
}

// ===== Setup =====
void setup() {
    Serial.begin(115200);
    Serial.println("\n ⚡ ...بورد الحساسات يبدأ");

    pinMode(PIR_PIN, INPUT);
    pirStartTime = millis();
    Serial.println("⌚ PIR (30 ثانية) ...يستقر");

    dht.begin();
    delay(2000);
    Serial.println("☑️ DHT11 + MQ-2 جاهزين");

    connectWiFi();
```

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```
}

// ===== Loop =====

void loop() {
    // إعادة الاتصال لو الشبكة قطعت
    if (WiFi.status() != WL_CONNECTED) connectWiFi();

    // استقرار PIR
    if (!pirReady && millis() - pirStartTime >= pirWarmupTime) {
        pirReady = true;
        Serial.println("☑ PIR جاهز!");
    }

    // قراءة PIR
    motionDetected = pirReady && (digitalRead(PIR_PIN) == HIGH);

    // قراءة DHT11
    if (millis() - lastDHTRead >= 2000) {
        lastDHTRead = millis();
        float t = dht.readTemperature();
        float h = dht.readHumidity();
        if (!isnan(t) && !isnan(h)) {
            temperature = t;
            humidity = h;
            dhtFailCount = 0;

            Serial.println("🌡 " + String(temperature, 1) + "°C | 💧 " +
                String(humidity, 1) + "%");
        } else {
```

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```
dhtFailCount++;

if (dhtFailCount >= 5) Serial.println(" ⚠️ فشل متكرر في DHT11 - تحقق من التوصيل");
}
}

// قراءة MQ-2
gasValue = analogRead(GAS_PIN);

// إرسال كل 2 ثانية
if (millis() - lastSendTime >= sendInterval) {
    lastSendTime = millis();
    sendDataToMain();
}

delay(200);
}
```